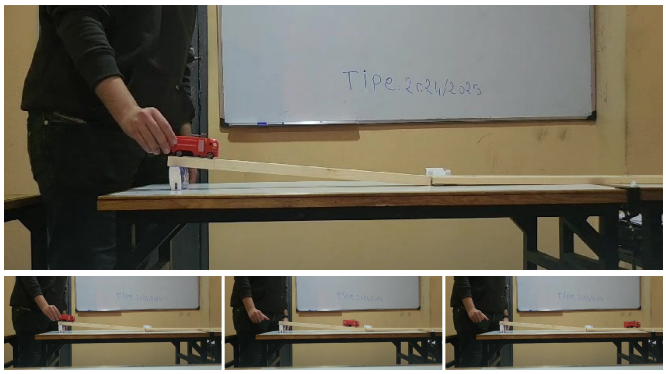
Flywheel rotor design

From the inertia-optimal ring to a stress-limited, printable profile

Hamza Houir Alami · Project grown from my TIPE research project (classes préparatoires, 2024–2025) · Python (SciPy, scikit-fem), CalculiX / Abaqus format, Fusion 360, Tracker, Arduino



Me running the TIPE braking experiment (Feb. 2025): a 34 g truck rolls down a ramp and is tracked frame by frame with Tracker.

QUESTION

Can a flywheel store part of the energy a car loses when braking, without degrading braking? Two sub-questions: *which shape stores the most energy*, and *what happens to braking*.

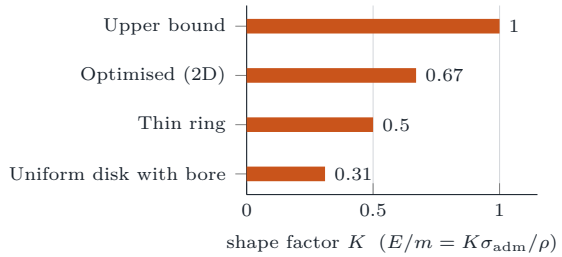
WHAT I FOUND

- **Shape at fixed speed.** The circle (fixed perimeter) and the outer annulus (fixed area, bounded radius) maximise inertia. The circle was found numerically in the TIPE; both results were then **proved** (Wirtinger inequality, radial rearrangement).
- **The missing constraint.** A real flywheel is limited by material strength, not speed. With a stress limit, the best shape becomes a **profiled disk**: shape factor 0.57–0.67 in 2D, against 0.5 for a thin ring.
- **1D optima fail in 2D.** Their near-vertical hub steps create a stress singularity, confirmed with CalculiX. A slope limit fixes it.
- **Braking.** The deceleration model matches the measurement ($f \approx 0.09$–0.10). Re-analysing the data showed the flywheel effect was not established: the two runs started with 28 to 47 % different kinetic energy.

HOW THE PROJECT EVOLVED

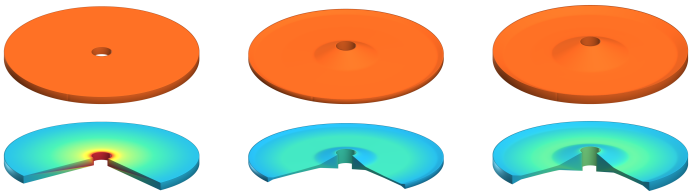


ENERGY PER KG BEFORE FAILURE



The TIPE optimum (thin ring) is beaten once strength is the limit. Optimised: 0.57 (2 mm web) to 0.67 (1 mm web).

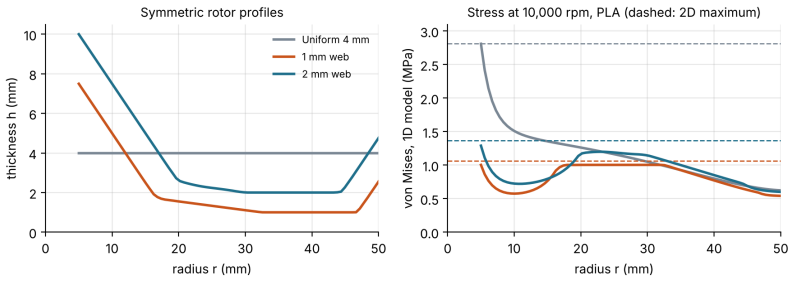
THE THREE PRINTABLE ROTORS



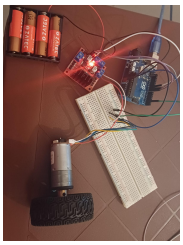
Uniform reference, 1 mm web, 2 mm web ($b = 50$ mm). Bottom: von Mises stress at 10,000 rpm in PLA (1D model).

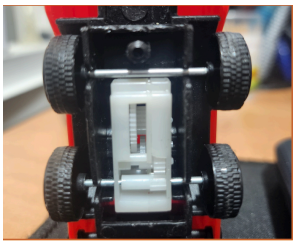
RIGOUR

Abaqus input files, solved with CalculiX, agree with my 2D FE code to 0.4–0.8 %. 2D results stable over five meshes (under 0.3 % change between the two finest). Trifilar pendulum designed to measure inertia: it separates the two profiles ($j = 0.420$ vs 0.462) by about 9 standard uncertainties.



Profiles of the three symmetric rotors and computed stress along the radius.





Left: first Arduino test bench (motor, encoder, wheel). Right: torsion spring and gear train used to mimic a flywheel on the truck.

DESIGN FOR PRINTING

Parts printed flat, without supports: web of at least 1 mm, slope ≤ 0.5. The choice of web depends on the printed thickness error: beyond about 0.13 mm, the 2 mm web wins.